Chapter 25: Problems with Solutions

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1. Oppositely charged parallel plates are separated by 5.33 mm. A potential difference of 600 V exists between the plates. (a) What is the magnitude of the electric field between the plates? (b) What is the magnitude of the force on an electron between the plates? (c) How much work must be done on the electron to move it to the negative plate if it is initially positioned 2.90 mm from the positive plate?

(a) From Equation 25.6,

$$E = \frac{|\Delta V|}{d} = \frac{600\ \text{J/C}}{5.33\times10^{-3}\ \text{m}} = \boxed{1.13\times10^{5}\ \text{N/C}}$$

(b) The force on an electron is given by

$$F = |q|E = (1.60\times10^{-19}\ \text{C})(1.13\times10^{5}\ \text{N/C}) = \boxed{1.80\times10^{-14}\ \text{N}}$$

(c) Because the electron is repelled by the negative plate, the force used to move the electron must be applied in the direction of the electron's displacement. The work done to move the electron is

$$W = F\cdot s\cos\theta = (1.80\times10^{-14}\ \text{N})[(5.33-2.00)\times10^{-3}\ \text{m}]\cos0°$$

$$= \boxed{4.37\times10^{-17}\ \text{J}}$$

3. (a) Calculate the speed of a proton that is accelerated **M** from rest through an electric potential difference of 120 V. (b) Calculate the speed of an electron that is accelerated through the same electric potential difference.

(a) Energy of the proton-field system is conserved as the proton moves from high to low potential, which can be defined for this problem as moving from 120 V down to 0 V.

$$K_i + U_i = K_f + U_f: \qquad 0 + qV = \frac{1}{2}mv_p^2 + 0$$

$$(1.60\times10^{-19}\ \text{C})(120\ \text{V})\left(\frac{1\ \text{J}}{1\ \text{V}\cdot\text{C}}\right) = \frac{1}{2}(1.67\times10^{-27}\ \text{kg})v_p^2$$

$$v_p = \boxed{1.52\times10^{5}\ \text{m/s}}$$

(b) The electron will gain speed in moving the other way,

from $V_i = 0$ to $V_f = 120$ V: $K_i + U_i = K_f + U_f$

$$0 + 0 = \frac{1}{2}mv_e^2 + qV$$

$$0 = \frac{1}{2}(9.11\times10^{-31}\ \text{kg})v_e^2 + (-1.60\times10^{-19}\ \text{C})(120\ \text{J/C})$$

$$v_e = \boxed{6.49\times10^{6}\ \text{m/s}}$$

5. A uniform electric field **W** of magnitude 325 V/m is directed in the negative y direction in Figure P25.5. The coordinates of point Ⓐ are (−0.200, −0.300) m, and those of point Ⓑ are (0.400, 0.500) m. Calculate the electric potential difference $V_B - V_A$ using the dashed-line path.

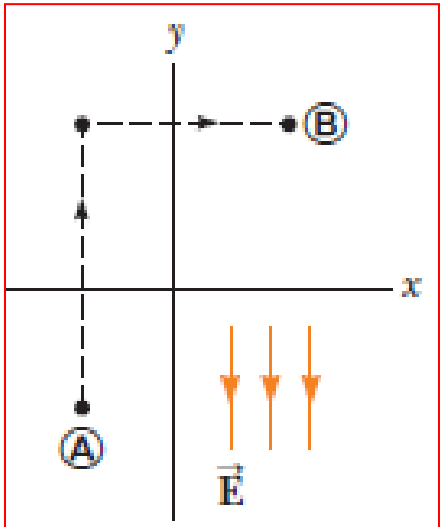


Figure P25.5

The electric field is uniform. By Equation 25.3,

$$V_B - V_A = -\int_A^B \vec{E}\cdot d\vec{s} = -\int_A^C \vec{E}\cdot d\vec{s} - \int_C^B \vec{E}\cdot d\vec{s}$$

$$V_B - V_A = (-E\cos 180°)\int_{-0.300}^{0.500} dy - (E\cos 90.0°)\int_{-0.200}^{0.400} dx$$

$$V_B - V_A = (325\ \text{V/m})(0.800\ \text{m}) = \boxed{+260\ \text{V}}$$

7. AMT M An electron moving parallel to the x axis has an initial speed of 3.70×10^6 m/s at the origin. Its speed is reduced to 1.40×10^5 m/s at the point $x = 2.00$ cm. (a) Calculate the electric potential difference between the origin and that point. (b) Which point is at the higher potential?

We use the energy version of the isolated system model to equate the energy of the electron-field system when the electron is at $x = 0$ to the energy when the electron is at $x = 2.00$ cm. The unknown will be the difference in potential $V_f - V_i$. Thus, $K_i + U_i = K_f + U_f$ becomes

$$\frac{1}{2}mv_i^2 + qV_i = \frac{1}{2}mv_f^2 + qV_f$$

or

$$\frac{1}{2}m\left(v_i^2 - v_f^2\right) = q\left(V_f - V_i\right),$$

so

$$V_f - V_i = \Delta V = \frac{m\left(v_i^2 - v_f^2\right)}{2q}.$$

(a) Noting that the electron's charge is negative, and evaluating the potential difference, we have

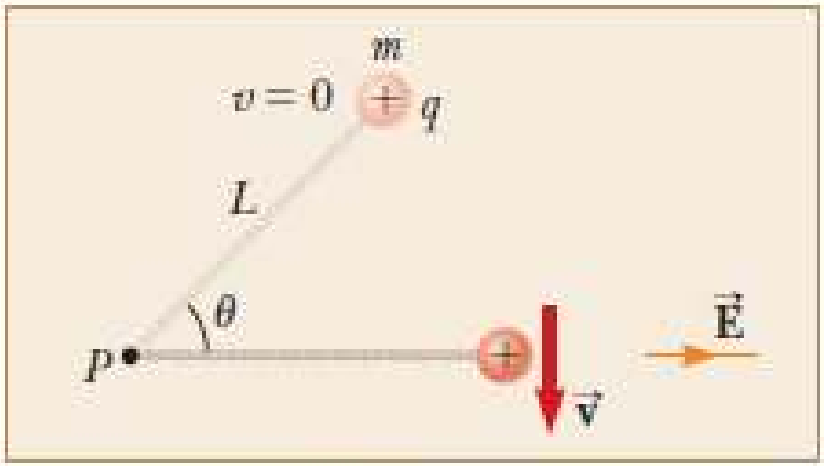
$$\Delta V = \frac{\left(9.11\times10^{-31}\ \text{kg}\right)\left[(3.70\times10^{6}\ \text{m/s})^2 - (1.40\times10^{5}\ \text{m/s})^2\right]}{2\left(-1.60\times10^{-19}\ \text{C}\right)}$$

$$= \boxed{-38.9\ \text{V}}$$

(b) The negative sign means that the 2.00-cm location is lower in potential than the origin:

The origin is at the higher potential.

9. AMT A particle having charge $q = +2.00\ \mu C$ and mass $m = 0.010\ 0$ kg is connected to a string that is $L = 1.50$ m long and tied to the pivot point P in Figure P25.9. The particle, string, and pivot point all lie on a frictionless,



Top view

Figure P25.9

horizontal table. The particle is released from rest when the string makes an angle $\theta = 60.0°$ with a uniform electric field of magnitude $E = 300$ V/m. Determine the speed of the particle when the string is parallel to the electric field.

Arbitrarily take $V = 0$ at point P. Then the potential at the original position of the charge is (by Equation 25.3)

$$\Delta V = V - 0 = V = -\vec{E} \cdot \vec{s} = -EL\cos\theta \quad \text{(relative to } P\text{)}$$

At the final point a,

$$V = -EL \qquad \text{(relative to } P\text{)}$$

Because the table is frictionless and the particle-field system is isolated, we have

$$(K + U)_i = (K + U)_f$$

or
$$0 - qEL\cos\theta = \frac{1}{2}mv^2 - qEL$$

solving for the speed gives

$$v = \sqrt{\frac{2qEL(1-\cos\theta)}{m}}$$

$$= \sqrt{\frac{2\left(2.00\times10^{-6}\ \text{C}\right)(300\ \text{N/C})(1.50\ \text{m})(1-\cos 60.0^\circ)}{0.010\,0\ \text{kg}}}$$

$$= \boxed{0.300\ \text{m/s}}$$

14. The two charges in Figure P25.14 are separated by $d = 2.00$ cm. Find the electric potential at (a) point A and (b) point B, which is halfway between the charges.

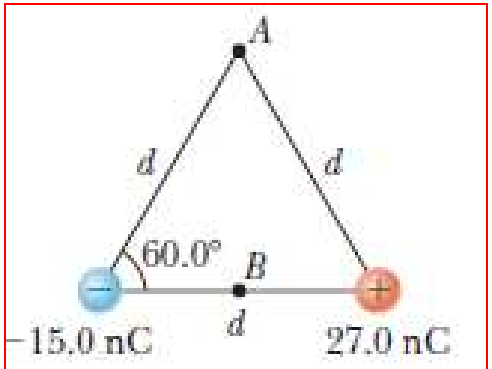


Figure P25.14

The potential due to the two charges is given by $V = k_e \sum_i \frac{q_i}{r_i}$.

(a) The electric potential at point A is

$$V = k_e \sum_i \frac{q_i}{r_i} = \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right) \times \left(\frac{-15.0 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} + \frac{27.0 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right)$$

$$= \boxed{5.39 \text{ kV}}$$

(b) The electric potential at point B is

$$V = k_e \sum_i \frac{q_i}{r_i} = \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right) \times \left(\frac{-15.0 \times 10^{-9} \text{ C}}{1.00 \times 10^{-2} \text{ m}} + \frac{27.0 \times 10^{-9} \text{ C}}{1.00 \times 10^{-2} \text{ m}} \right)$$

$$= \boxed{10.8 \text{ kV}}$$

15. Three positive charges are located at the corners of an equilateral triangle as in Figure P25.15. Find an expression for the electric potential at the center of the triangle.

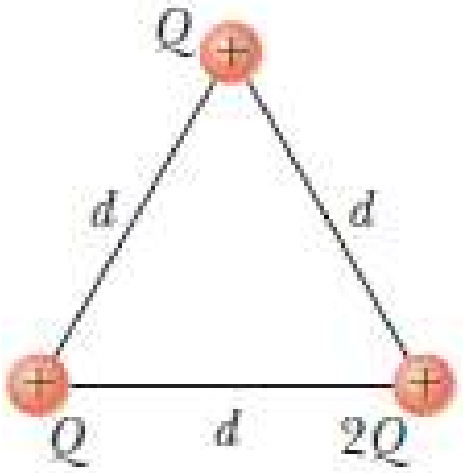
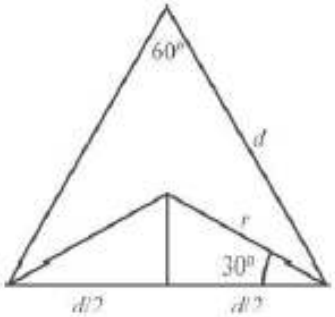


Figure P25.15

By symmetry, a line from the center to each vertex forms a 30° angle with each side of the triangle. The figure shows the relationship between the length d of a side of the equilateral triangle and the distance r from a vertex to the center:



ANS. FIG. P25.15

$$r\cos 30.0^\circ = d/2$$

$$\rightarrow r = d/(2\cos 30.0^\circ)$$

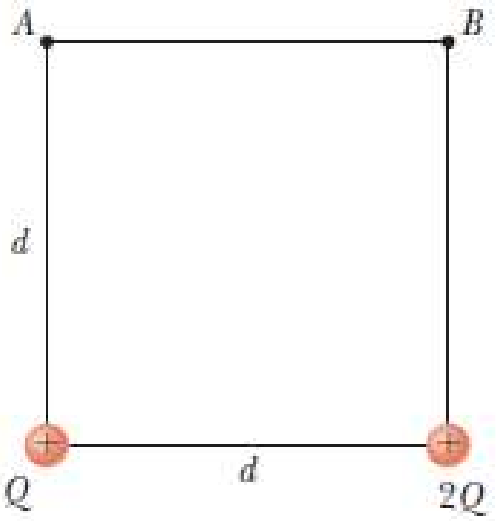
The electric potential at the center is

$$V = k_e \sum_i \frac{q_i}{r_i}$$

$$= k_e \left(\frac{Q}{d/(2\cos 30.0^\circ)} + \frac{Q}{d/(2\cos 30.0^\circ)} + \frac{2Q}{d/(2\cos 30.0^\circ)} \right)$$

$$V = (4)\left(2\cos 30.0^\circ \, k_e \frac{Q}{d} \right) = \boxed{6.93 k_e \frac{Q}{d}}$$

18. The two charges in Figure P25.18 are separated by a distance $d = 2.00$ cm, and $Q = +5.00$ nC. Find (a) the electric potential at A, (b) the electric potential at B, and (c) the electric potential difference between B and A.



(a) $$V_A = k_e \sum_i \frac{q_i}{r_i} = k_e \left(\frac{Q}{d} + \frac{2Q}{d\sqrt{2}} \right) = k_e \frac{Q}{d} \left(1 + \sqrt{2} \right)$$

$$V = \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2 \right) \left(\frac{5.00 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right) \left(1 + \sqrt{2} \right) = \boxed{5.43 \text{ kV}}$$

(b) $$V_B = k_e \sum_i \frac{q_i}{r_i} = k_e \left(\frac{Q}{d\sqrt{2}} + \frac{2Q}{d} \right) = k_e \frac{Q}{d} \left(\frac{1}{\sqrt{2}} + 2 \right)$$

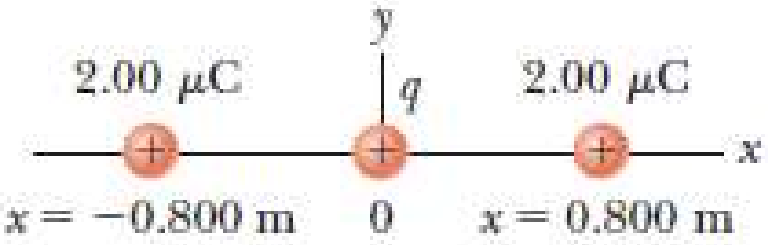
$$V_B = \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2 \right) \left(\frac{5.00 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right) \left(\frac{1}{\sqrt{2}} + 2 \right) = \boxed{6.08 \text{ kV}}$$

(c) $$V_B - V_A = k_e \frac{Q}{d} \left(\frac{1}{\sqrt{2}} + 2 \right) - k_e \frac{Q}{d} \left(1 + \sqrt{2} \right) = k_e \frac{Q}{d} \left(\frac{1}{\sqrt{2}} + 1 - \sqrt{2} \right)$$

$$V_B - V_A = \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2 \right) \left(\frac{5.00 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right) \left(\frac{1}{\sqrt{2}} + 1 - \sqrt{2} \right)$$

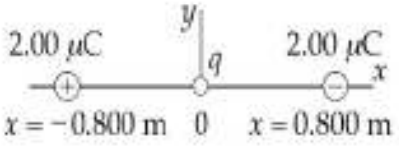
$$= \boxed{658 \text{ V}}$$

19. Given two particles with 2.00-μC charges as shown in Figure P25.19 and a particle with charge $q = 1.28 \times 10^{-18}$ C at the origin, (a) what is the net force exerted by the two 2.00-μC charges on the charge q? (b) What is the electric field at the origin due to the two 2.00-μC particles? (c) What is the electric potential at the origin due to the two 2.00-μC particles?



(a) Since the charges are equal and placed symmetrically, $\boxed{F = 0}$.

(b) Since $F = qE = 0$, $\boxed{E = 0}$.



ANS. FIG. P25.19

(c) $$V = 2k_e \frac{q}{r}$$

$$= 2\left(8.99 \times 10^9 \ \text{N} \cdot \text{m}^2/\text{C}^2\right)\left(\frac{2.00 \times 10^{-6} \ \text{C}}{0.800 \ \text{m}}\right)$$

$$V = 4.50 \times 10^4 \ \text{V} = \boxed{45.0 \ \text{kV}}$$

27. Four identical charged particles ($q = +10.0\ \mu C$) are located on the corners of a rectangle as shown in Figure P25.27. The dimensions of the rectangle are $L = 60.0$ cm and $W = 15.0$ cm. Calculate the change in electric potential energy of the system as the particle at the lower left corner in Figure P25.27 is brought to this position from infinitely far away. Assume the other three particles in Figure P25.27 remain fixed in position.

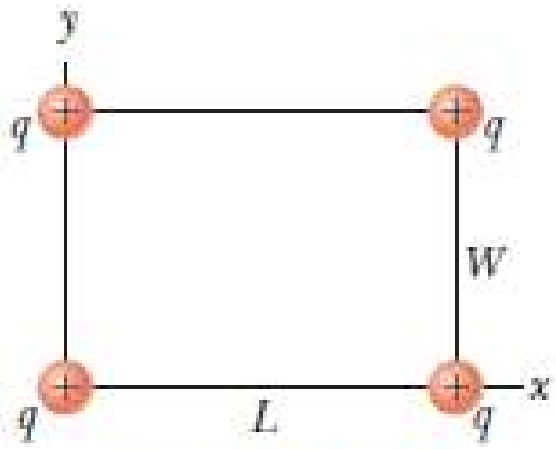


Figure P25.27

The total change in potential energy is the sum of the change in potential energy of the $q_1 - q_4$, $q_2 - q_4$, and $q_3 - q_4$ particle systems:

$$U_e = q_4V_1 + q_4V_2 + q_4V_3 = q_4k_e\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3}\right)$$

$$U_e = \left(10.0\times10^{-6}\ \text{C}\right)^2\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)$$

$$\times\left(\frac{1}{0.600\ \text{m}} + \frac{1}{0.150\ \text{m}} + \frac{1}{\sqrt{(0.600\ \text{m})^2 + (0.150\ \text{m})^2}}\right)$$

$$U_e = \boxed{8.95\ \text{J}}$$

36. Figure P25.36 represents a graph of the electric potential in a region of space versus position x, where the electric field is parallel to the x axis. Draw a graph of the x component of the electric field versus x in this region.

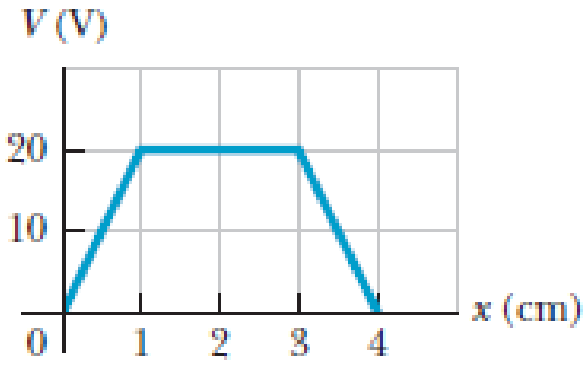


Figure P25.36

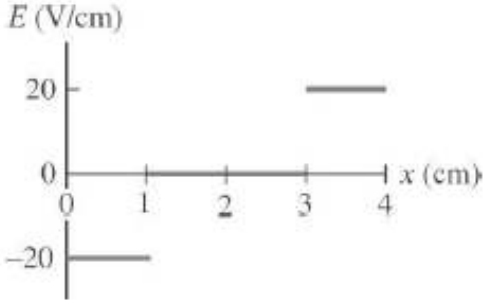
$$E_x = -\frac{\partial V}{\partial x} \quad \rightarrow \quad E_x = -\frac{\Delta V}{\Delta x} = -\text{(slope of line)}$$

The sign indicates the direction of the x component of the field.

$x = 0$ to 1 cm: $$E_x = -\frac{\Delta V}{\Delta x} = -\frac{20\text{ V} - 0}{1\text{ cm}} = -20\text{ V/cm}$$

$x = 1$ to 3 cm: $$E_x = -\frac{\Delta V}{\Delta x} = -\frac{0}{2\text{ cm}} = 0\text{ V/m}$$

$x = 3$ to 4 cm: $$E_x = -\frac{\Delta V}{\Delta x} = -\frac{0 - 20\text{ V}}{1\text{ cm}} = +20\text{ V/cm}$$



39. W Over a certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. (a) Find the expressions for the x, y, and z components of the electric field over this region. (b) What is the magnitude of the field at the point P that has coordinates (1.00, 0, −2.00) m?

(a) $V = 5x - 3x^2y + 2yz^2$, where x, y and z are in meters and V is in volts.

$$E_x = -\frac{\partial V}{\partial x} = -5 + 6xy$$

$$E_y = -\frac{\partial V}{\partial y} = +3x^2 - 2z^2$$

$$E_z = -\frac{\partial V}{\partial z} = -4yz$$

which gives

$$\boxed{\vec{E} = (-5 + 6xy)\hat{i} + (3x^2 - 2z^2)\hat{j} - 4yz\hat{k}}$$

(b) Evaluate E at (1.00, 0, −2.00) m, suppressing units,

$$E_x = -5 + 6(1.00)(0) = -5.00$$

$$E_y = 3(1.00)^2 - 2(-2.00)^2 = -5.00$$

$$E_z = -4(0)(-2.00) = 0$$

which gives

$$E = \sqrt{E_x^2 + E_y^2 + E_z^2} = \sqrt{(-5.00)^2 + (-5.00)^2 + 0^2} = \boxed{7.07 \text{ N/C}}$$

41. The electric potential inside a charged spherical conductor of radius R is given by $V = k_e Q/R$, and the potential outside is given by $V = k_e Q/r$. Using $E_r = -dV/dr$, derive the electric field (a) inside and (b) outside this charge distribution.

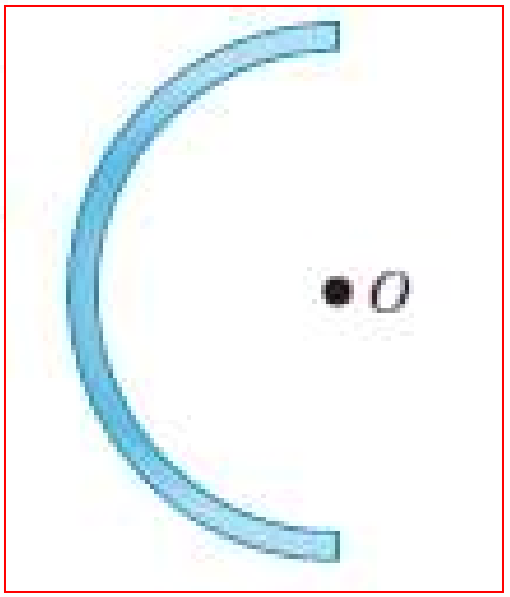
(a) For $r < R$, $$V = \frac{k_e Q}{R}$$

$$E_r = -\frac{dV}{dr} = \boxed{0}$$

(b) For $r \geq R$, $$V = \frac{k_e Q}{r}$$

$$E_r = -\frac{dV}{dr} = -\left(-\frac{k_e Q}{r^2}\right) = \boxed{\frac{k_e Q}{r^2}}$$

44. A uniformly charged insulating rod of length 14.0 cm is bent into the shape of a semicircle as shown in Figure P25.44. The rod has a total charge of $-7.50\ \mu C$. Find the electric potential at O, the center of the semicircle.



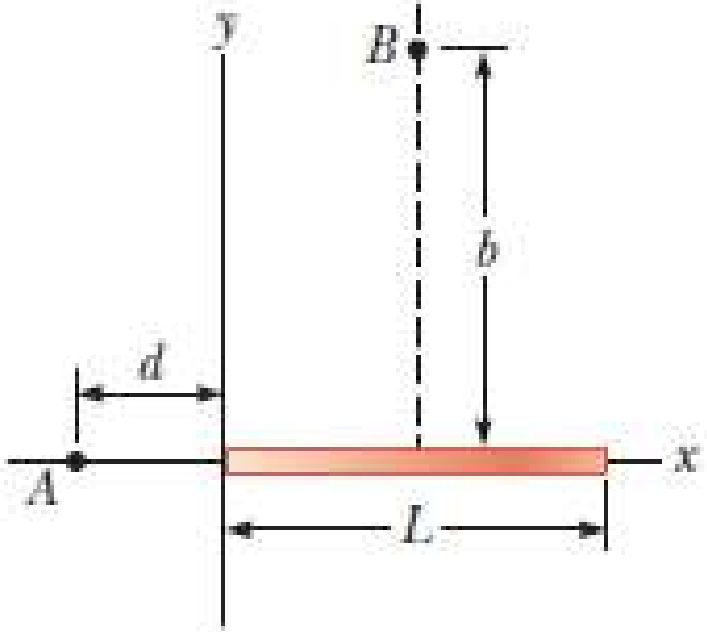
$$V = \int dV = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r}$$

All bits of charge are at the same distance from O. So

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{Q}{R}\right) = \left(8.99\times10^{9}\ \mathrm{N\cdot m^2/C^2}\right)\left(\frac{-7.50\times10^{-6}\ \mathrm{C}}{0.140\ \mathrm{m}/\pi}\right)$$

$$= \boxed{-1.51\ \mathrm{MV}}$$

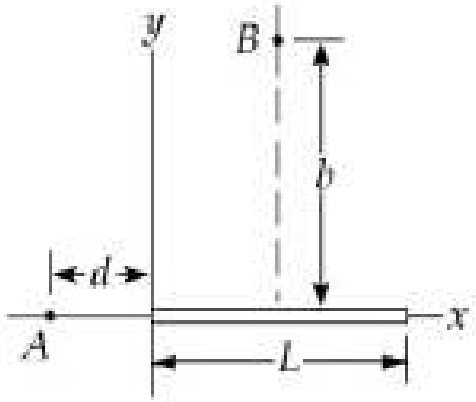
45. A rod of length L (Fig. P25.45) lies along the x axis with its left end at the origin. It has a nonuniform charge



density $\lambda = \alpha x$, where α is a positive constant. (a) What are the units of α? (b) Calculate the electric potential at A.

(a) As a linear charge density, λ has units of C/m. So $\alpha = \lambda/x$ must have units of C/m^2:

$$[\alpha] = \left[\frac{\lambda}{x}\right] = \frac{C}{m} \cdot \left(\frac{1}{m}\right) = \boxed{\frac{C}{m^2}}$$



ANS. FIG. P25.45

(b) Consider a small segment of the rod at location x and of length dx. The amount of charge on it is $\lambda\, dx = (\alpha x)\, dx$. Its distance from A is $d + x$, so its contribution to the electric potential at A is

$$dV = k_e \frac{dq}{r} = k_e \frac{\alpha x\, dx}{d + x}$$

Relative to $V = 0$ infinitely far away, to find the potential at A we must integrate these contributions for the whole rod, from $x = 0$ to $x = L$. Then $V = \int_{\text{all q}} dV = \int_0^L \frac{k_e \alpha x}{d + x}\, dx$.

To perform the integral, make a change of variables to $u = d + x$, $du = dx$, $u(\text{at } x = 0) = d$, and $u(\text{at } x = L) = d + L$:

$$V = \int_d^{d+L} \frac{k_e \alpha (u - d)}{u} du = k_e \alpha \int_d^{d+L} du - k_e \alpha d \int_d^{d+L} \left(\frac{1}{u}\right) du$$

$$V = k_e \alpha u \Big|_d^{d+L} - k_e \alpha d \ln u \Big|_d^{d+L}$$

$$= k_e \alpha (d + L - d) - k_e \alpha d \left[\ln(d + L) - \ln d\right]$$

$$\boxed{V = k_e \alpha \left[L - d \ln\left(1 + \frac{L}{d}\right)\right]}$$

47. A wire having a uniform linear charge density λ is bent into the shape shown in Figure P25.47. Find the electric potential at point O.

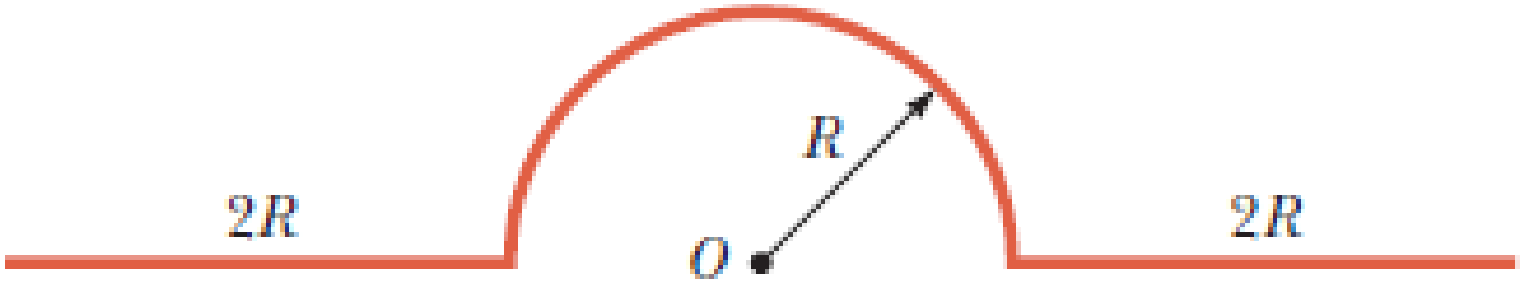


Figure P25.47

$$V = k_e \int_{\text{all charge}} \frac{dq}{r} = k_e \int_{-3R}^{-R} \frac{\lambda dx}{-x} + k_e \int_{\text{semicircle}} \frac{\lambda ds}{R} + k_e \int_{R}^{3R} \frac{\lambda dx}{x}$$

$$V = -k_e \lambda \ln(-x)\Big|_{-3R}^{-R} + \frac{k_e \lambda}{R} \pi R + k_e \lambda \ln x \Big|_{R}^{3R}$$

$$V = k_e \lambda \ln \frac{3R}{R} + k_e \lambda \pi + k_e \lambda \ln 3 = \boxed{k_e \lambda (\pi + 2 \ln 3)}$$

49. How many electrons should be removed from an initially uncharged spherical conductor of radius 0.300 m to produce a potential of 7.50 kV at the surface?

Substituting given values into $V = \frac{k_e Q}{r}$, with $Q = Nq$:

$$7.50 \times 10^3 \text{ V} = \frac{\left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right) Q}{0.300 \text{ m}}$$

Substituting $q = 2.50 \times 10^{-7}$ C,

$$N = \frac{2.50 \times 10^{-7} \text{ C}}{1.60 \times 10^{-19} \text{ C/e}^-} = \boxed{1.56 \times 10^{12} \text{ electrons}}$$

50. M A spherical conductor has a radius of 14.0 cm and a charge of 26.0 μC. Calculate the electric field and the electric potential at (a) $r = 10.0$ cm, (b) $r = 20.0$ cm, and (c) $r = 14.0$ cm from the center.

For points on the surface and outside, the sphere of charge behaves like a charged particle at its center, both for creating field and potential.

(a) Inside a conductor when charges are not moving, the electric field is zero and the potential is uniform, the same as on the surface, and $E = \boxed{0}$.

$$V = \frac{k_e q}{R} = \frac{(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2)(26.0\times10^{-6}\ \text{C})}{0.140\ \text{m}} = \boxed{1.67\ \text{MV}}$$

(b) $$E = \frac{k_e q}{r^2} = \frac{(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2)(26.0\times10^{-6}\ \text{C})}{(0.200\ \text{m})^2}$$

$= \boxed{5.84\ \text{MN/C}}$ away

$$V = \frac{k_e q}{R} = \frac{(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2)(26.0\times10^{-6}\ \text{C})}{0.200\ \text{m}} = \boxed{1.17\ \text{MV}}$$

(c) $$E = \frac{k_e q}{R^2} = \frac{(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2)(26.0\times10^{-6}\ \text{C})}{(0.140\ \text{m})^2}$$

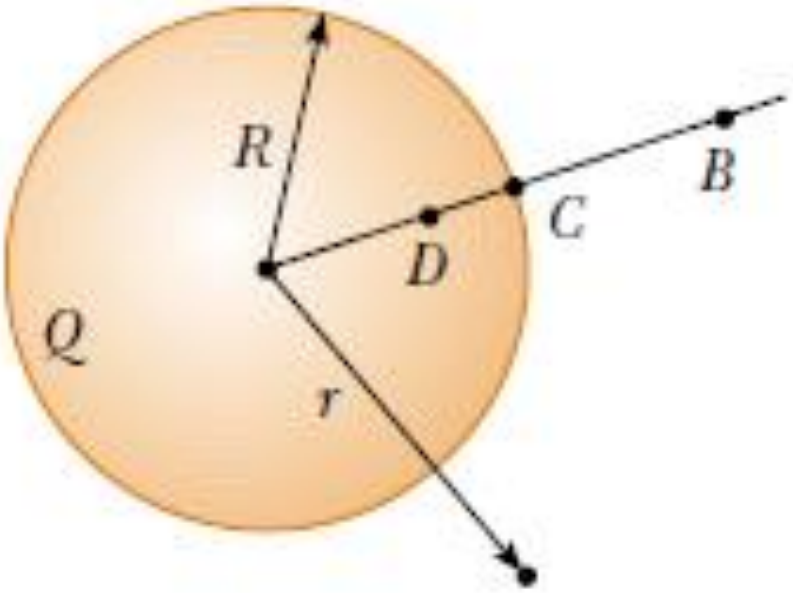
$= \boxed{11.9\ \text{MN/C}}$ away

$$V = \frac{k_e q}{R} = \boxed{1.67\ \text{MV}}$$

Electric Potential Due to a Uniformly Charged Sphere

An insulating solid sphere of radius R has a uniform positive volume charge density and total charge Q.

(A) Find the electric potential at a point outside the sphere, that is, for $r > R$. Take the potential to be zero at $r = \infty$.



Solution In Example 24.5, we found that the magnitude of the electric field outside a uniformly charged sphere of radius R is

$$E_r = k_e \frac{Q}{r^2} \qquad (\text{for } r > R)$$

where the field is directed radially outward when Q is positive. This is the same as the field due to a point charge, which we studied in Section 23.4. In this case, to obtain the electric potential at an exterior point, such as B in Figure 25.19, we use Equation 25.10, choosing point A as $r = \infty$:

$$V_B - V_A = k_e Q \left[\frac{1}{r_B} - \frac{1}{r_A} \right]$$

$$V_B - 0 = k_e Q \left[\frac{1}{r_B} - 0 \right]$$

$$V_B = k_e \frac{Q}{r} \qquad (\text{for } r > R)$$

Because the potential must be continuous at $r = R$, we can use this expression to obtain the potential at the surface of the sphere. That is, the potential at a point such as C shown in Figure 25.19 is

$$V_C = k_e \frac{Q}{R} \qquad (\text{for } r = R)$$

(B) Find the potential at a point inside the sphere, that is, for $r < R$.

Solution In Example 24.5 we found that the electric field inside an insulating uniformly charged sphere is

$$E_r = \frac{k_e Q}{R^3} r \qquad (\text{for } r < R)$$

We can use this result and Equation 25.3 to evaluate the potential difference $V_D - V_C$ at some interior point D:

$$V_D - V_C = -\int_R^r E_r \, dr = -\frac{k_e Q}{R^3} \int_R^r r \, dr = \frac{k_e Q}{2R^3} (R^2 - r^2)$$

Substituting $V_C = k_e Q / R$ into this expression and solving for V_D, we obtain

$$V_D = \frac{k_e Q}{2R} \left(3 - \frac{r^2}{R^2} \right) \qquad (\text{for } r < R) \quad (25.26)$$